What exactness costs: *O(n) drawers vs an O(1) whiteboard*

Topic: Key–Value Caching, Limitations, and Alternate Approaches · Live: `mbahc25088.github.io/IITKGP` · Repo: `github.com/mbahc25088/IITKGP`

OUR BET — BREAK IT

KV-cache grows one slot per token, O(n), always exact. Fixed recurrent state stays O(1) but silently forgets on collision. Flood 500 pairs into 16 slots: if recall stays perfect, we lose.

Why this hurts now. Long context made exactness unaffordable — each new token attends over all old ones. Linear attention, SSMs, and recurrent states collapse the cache to a fixed matrix: O(T) compute, O(1) memory, paid in interference.

THE MECHANISM IN ONE PARAGRAPH

Per-token slots become additive state (BDH-CQ's additive-per-demonstration case). Our toys eat the same pairs (`mango-3 → 42`): cabinet = `Map` per pair; whiteboard = 5 flippable write rules (hash last-wins, mamba-gate, delta blend, hola +8 cache, bdh-ρ decay). At 120/32 the whiteboard already has ~80 overwrites; 500→16 craters each rule differently while KV stays 100%. Click a yellow cell and the table names the thieves. Truth next to estimate is the lesson.

WHO PAYS WHAT

Family	Memory	Recall	Bill	Good at
Transformer KV	O(n)	perfect	$O(T^2)$, OOMs	quality when you can pay
Mamba ’24	O(1) gated	weak far	cheap	long-stream throughput
HOLA ’26	O(1)+cache	recovered	cheap+cache	needle rescue
BDH / CQ	**O(1) synaptic**	sags loaded	lowest $/task seen	memory = reasoning

BDH WITHOUT THE MYSTIQUE

No drawers, just synapses written while reading: $\rho_{t-1}=\Sigma\ v^*x^TU^{t-\tau}$ (Kosowski et al., arXiv:2509.26507, Eqs.4–5; Hebbian 6–8; code App.E). BDH-GPU = ReLU low-rank + linear attention — not a Mamba SSM. BDH-CQ (arXiv:2608.09888, $S_t=U(S_{t-1},D_t)$) is the sibling Pareto point: 150M, 29.5% pass@2 ARC-AGI-1 at ~$0.0007/task, no chain-of-thought tokens. Mamba (Gu & Dao, 2024) is the baseline we compare against; HOLA (Cui, arXiv:2607.02303 — 340M, Wiki 27.32→22.92, RULER to 32K) states our claim back to us: O(1) at the price of lossy exact memory.

Labeled evidence. 134M BDH: 95%@32K → 82%@128K BABILong QA1–QA5 (pathway.com; vendor benchmark, pending contamination/independent/leaderboard review). Holds long, sags loaded — our toy's curve at scale.

Cost evidence. BDH-CQ ~11× cheaper/task than GPT-5.6 Luna Low (34.2% vs 29.5%) — one self-reported benchmark, weights unreleased; Pareto claim, not raw SOTA.

THE LIMIT WE WOULD DEFEND FIRST

Forgetting is silent — no flag — and session memory is not durable learning; fast-to-slow consolidation stays open. Our hash collisions cartoon superposition along shared key directions: same curve, simpler mechanism, disclosed. Next: BDH App.E code, HOLA §2–3, Equations of Reasoning posts.

Team-written with AI drafting help (Muse Spark) for code, prose, design; team traces, tests, defends all of it. Cloud 9.